

End-to-End E-Commerce Analytics Platform

Data Analysis Internship Project Report

Business Insights and Data-Driven Decision Making
Using PostgreSQL and Power BI

Covering Complete Project Analysis
(Data Modeling, SQL, Business Intelligence & Dashboard Development)

Dataset: Olist E-Commerce Dataset
Tools: PostgreSQL, SQL, Power BI, DAX, Data Modeling



Prepared as part of the Data Analysis Internship

By: BIMLESH KUMAR SAH

Email: sahbimlesh9@gmail.com

1. Introduction

The End-to-End E-Commerce Analytics Platform is a Business Intelligence and Data Analytics solution designed to transform raw e-commerce transaction data into meaningful business insights. In today's digital marketplace, e-commerce organizations generate massive volumes of data from customers, products, orders, payments, deliveries, and reviews. While this data contains valuable information, it is often stored in multiple tables and formats, making it difficult for business stakeholders to derive actionable insights.

This project was developed using PostgreSQL as the data storage and processing layer and Power BI as the reporting and visualization platform. The project utilizes the Olist E-Commerce Dataset, a real-world Brazilian e-commerce dataset containing information related to customers, sellers, products, orders, payments, logistics, and customer reviews.

The solution follows a structured analytics workflow in which raw data is imported into PostgreSQL, validated, optimized through indexing, transformed into business-friendly analytical views, and finally visualized using interactive Power BI dashboards. The platform enables stakeholders to monitor business performance, understand customer behavior, evaluate product and seller performance, track delivery efficiency, and measure customer satisfaction.

The primary goal of the project is to support data-driven decision-making by providing accurate, reliable, and visually intuitive reports that help businesses identify trends, opportunities, and operational challenges.

2. Problem Statement

Modern e-commerce businesses handle thousands of transactions every day. These transactions generate data across multiple business functions, including sales, inventory, logistics, customer engagement, and payments. Although organizations possess large amounts of data, extracting meaningful insights from raw transactional records remains a significant challenge.

Some common business problems faced by e-commerce organizations include:

- Lack of visibility into overall sales performance.
- Difficulty understanding customer purchasing behavior.
- Challenges in identifying high-performing and low-performing products.
- Limited ability to measure delivery efficiency and logistics performance.
- Inability to evaluate seller contribution to revenue generation.
- Difficulty monitoring customer satisfaction through ratings and reviews.
- Challenges in tracking payment preferences and purchasing patterns.

Without a centralized analytics system, decision-makers often rely on manual reporting processes, which are time-consuming, error-prone, and unable to provide real-time business visibility.

This project addresses these challenges by creating a centralized analytics platform that consolidates transactional data into a structured reporting model and provides actionable insights through interactive dashboards and business KPIs.

3. Objectives

Primary Objectives

The primary objective of this project is to develop a comprehensive analytics platform that enables organizations to monitor and evaluate key aspects of their e-commerce operations.

The specific objectives include:

- Analyzing sales performance across different time periods.
- Monitoring customer purchasing behavior and demographics.
- Tracking delivery performance and logistics efficiency.
- Evaluating seller contribution and performance metrics.
- Measuring customer satisfaction through reviews and ratings.
- Providing interactive dashboards for business users.
- Creating reusable analytical views for reporting purposes.

Secondary Objectives

In addition to operational reporting, the project also aims to support strategic business initiatives by:

- Improving business decision-making through data visualization.
- Identifying sales trends and growth opportunities.
- Supporting marketing and product strategies.
- Enhancing customer retention efforts through customer insights.
- Enabling performance benchmarking across products and sellers.
- Providing a scalable foundation for future predictive analytics initiatives.

4. Technology Stack

The project leverages industry-standard technologies for data storage, transformation, and visualization.

Layer	Technology
Database Management System	PostgreSQL
Query Language	SQL
Data Modeling	Star Schema Approach
Data Source	Olist E-Commerce Dataset
Business Intelligence Tool	Power BI
Data Transformation	SQL Views
Reporting Layer	Power BI Dashboards
Performance Optimization	PostgreSQL Indexing

PostgreSQL

PostgreSQL serves as the central data repository and analytical processing engine. It stores all transactional data and executes complex SQL queries used for reporting and analysis.

SQL

SQL is used for data validation, transformation, aggregation, and the creation of analytical views that support reporting requirements.

Power BI

Power BI provides interactive visualizations, KPI dashboards, trend analysis, and business reporting capabilities. DirectQuery mode is used to connect Power BI directly to PostgreSQL.

Star Schema Modeling

A star schema-inspired approach is implemented to improve reporting efficiency and simplify analytical queries.

5. System Workflow

The system follows a structured data analytics pipeline that transforms raw data into business insights.



Workflow Explanation

1. Raw CSV files are downloaded from Kaggle and imported into PostgreSQL.
2. Data quality checks are performed to validate data integrity.
3. Database indexes are created to improve query performance.
4. SQL views are developed to create business-friendly reporting structures.
5. Power BI connects to PostgreSQL using DirectQuery.
6. DAX measures are created for KPI calculations.
7. Interactive dashboards are developed to visualize business performance.
8. Stakeholders use these dashboards for analysis and decision-making.

6. Key Features

Sales Analytics

The platform provides detailed insights into sales performance through revenue analysis, order tracking, sales trends, and growth monitoring. Business users can identify seasonal patterns, monitor revenue fluctuations, and evaluate overall business performance.

Key Metrics:

- Total Revenue
- Total Orders
- Average Order Value (AOV)
- Monthly Revenue Trends
- Year-over-Year Growth

Customer Analytics

This helps businesses understand customer distribution, purchasing behavior, and geographic trends.

Key Metrics:

- Total Customers
- Customer Distribution by State
- Revenue by Customer Location
- Repeat Purchase Analysis

Product Analytics

This module evaluates product performance across categories and identifies products contributing most significantly to revenue generation.

Key Metrics:

- Revenue by Product Category
- Best-Selling Products
- Product Order Volume
- Product Performance Ranking

Seller Analytics

Seller analytics provides visibility into seller performance and contribution to overall business growth.

Key Metrics:

- Revenue by Seller
- Orders by Seller
- Seller Ranking & Seller Performance Comparison

Logistics Analytics

The logistics module measures delivery performance and identifies operational inefficiencies.

Key Metrics:

- Average Delivery Time
- On-Time Delivery Percentage
- Late Delivery Rate
- Delivery Performance Trends

Review Analytics

Customer review analysis helps businesses evaluate customer satisfaction and service quality.

Key Metrics:

- Average Review Rating
- Rating Distribution
- Positive vs Negative Reviews
- Customer Satisfaction Trends

Payment Analytics

Payment analytics provides insights into customer payment preferences and purchasing behavior.

Key Metrics:

- Revenue by Payment Type
- Installment Usage Analysis
- Payment Method Distribution

7. Results and Outcomes

The implementation of the End-to-End E-Commerce Analytics Platform successfully transformed raw transactional data into actionable business intelligence.

Key achievements include:

- Development of a centralized reporting platform.
- Creation of optimized analytical views in PostgreSQL.
- Implementation of interactive Power BI dashboards.
- Improved visibility into sales and operational performance.
- Enhanced understanding of customer purchasing behavior.
- Identification of high-performing products and sellers.
- Improved monitoring of logistics and delivery operations.
- Better evaluation of customer satisfaction metrics.

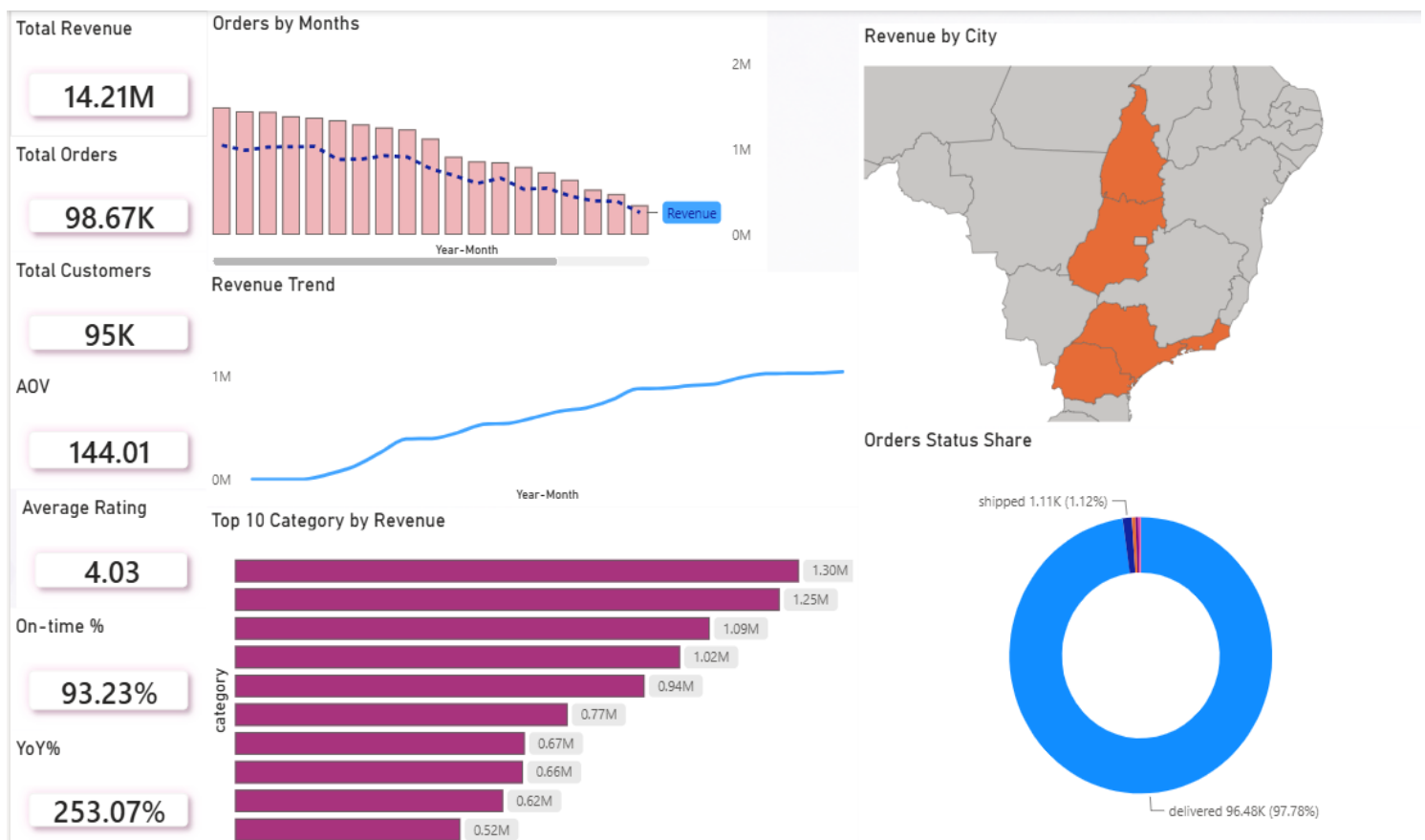


Fig1.1: Overview (Executive Dashboard)

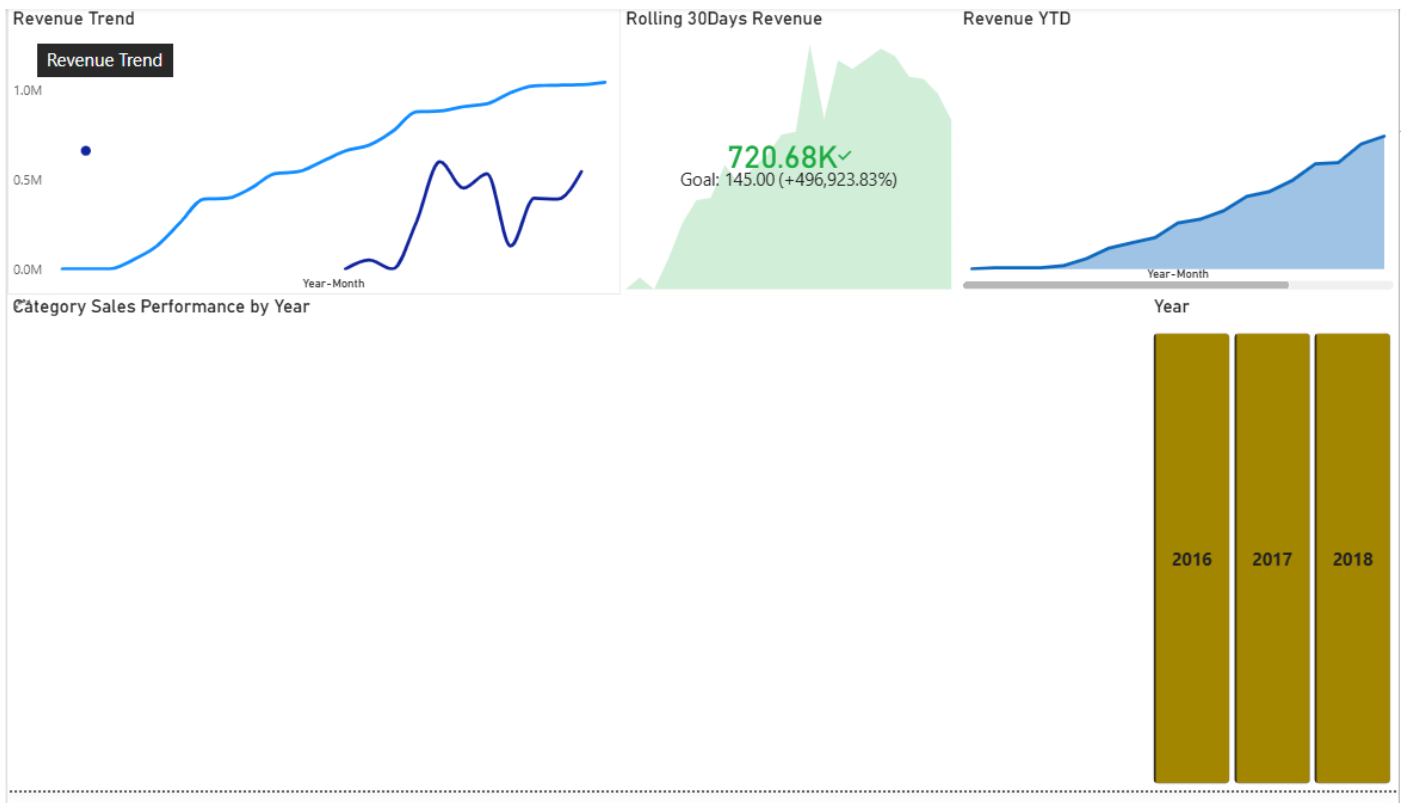


Fig1.2: Sales Trend Report

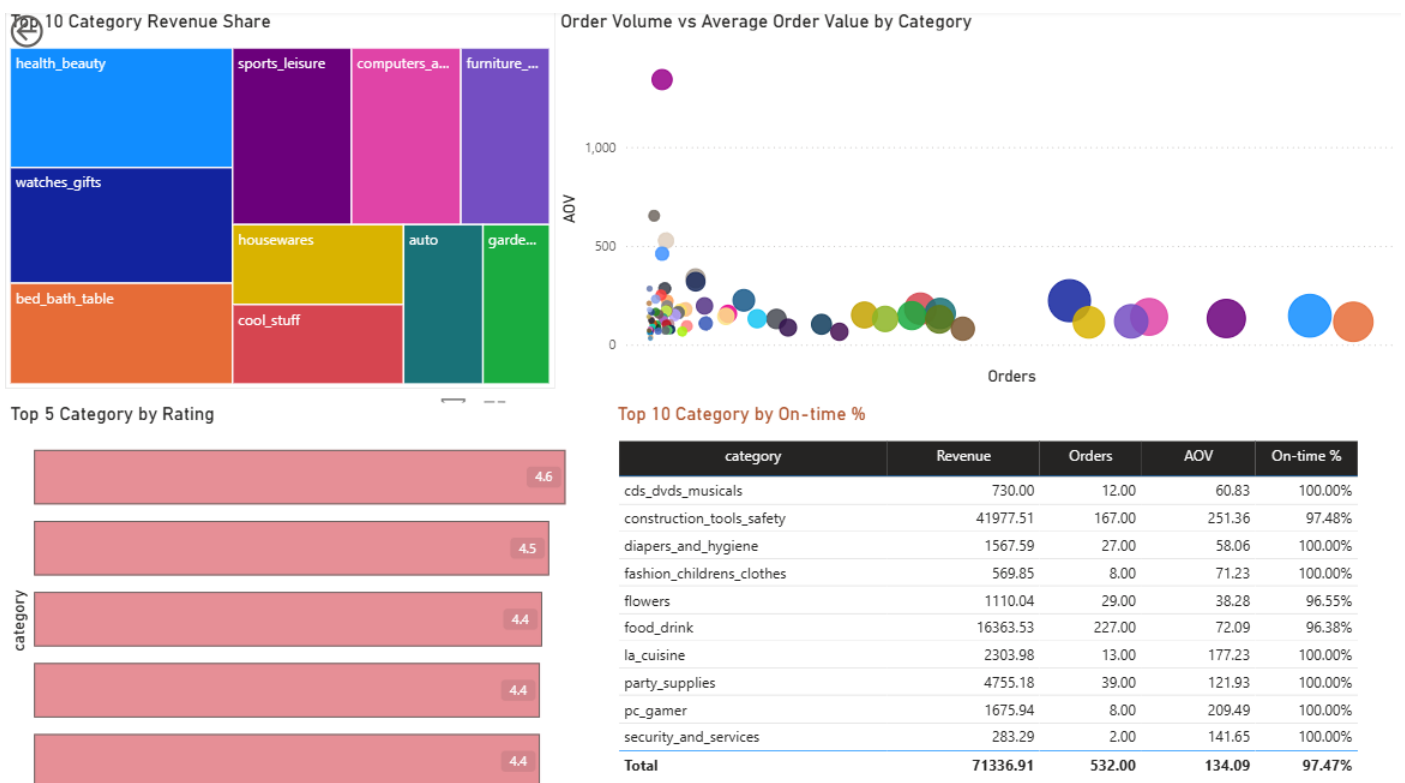


Fig1.3: Category and Product Report

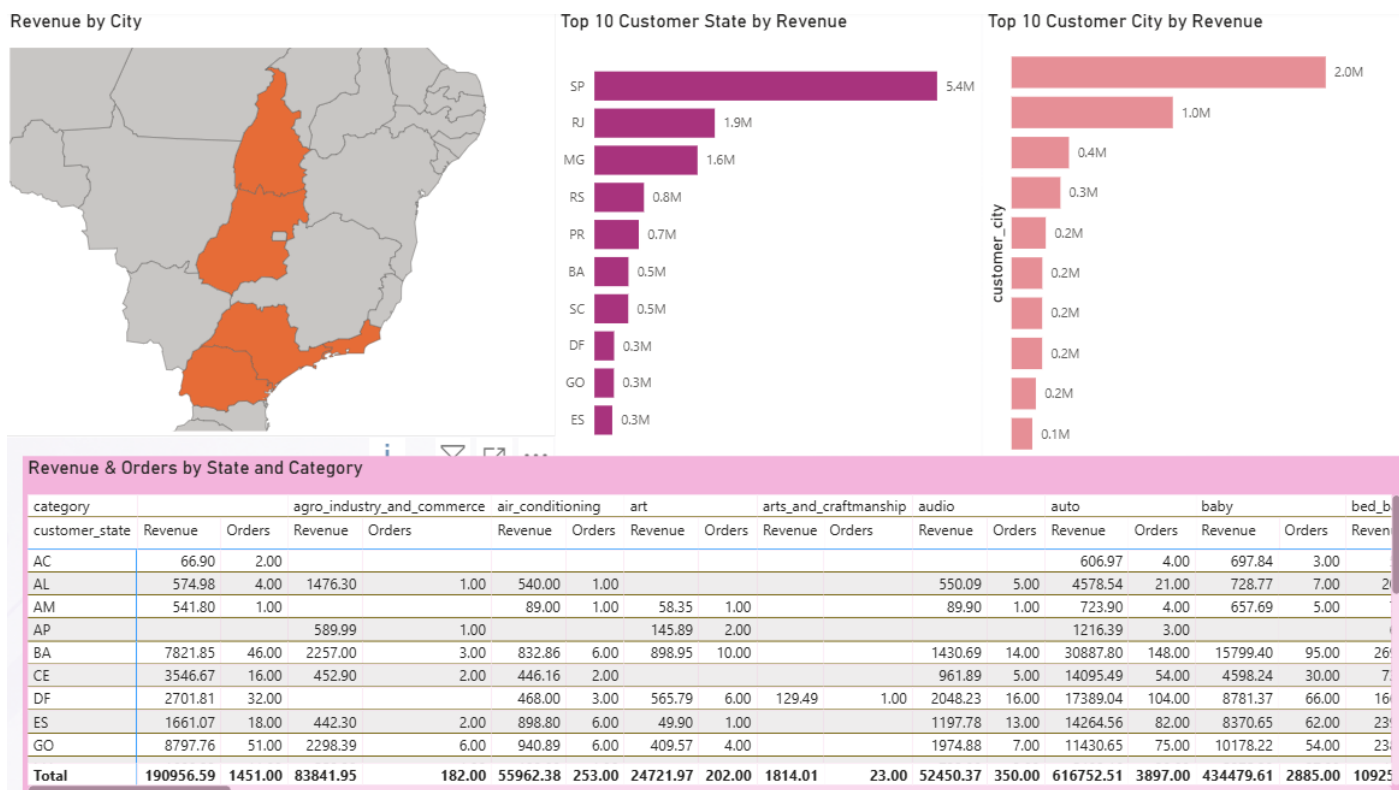


Fig1.4: Customer Report

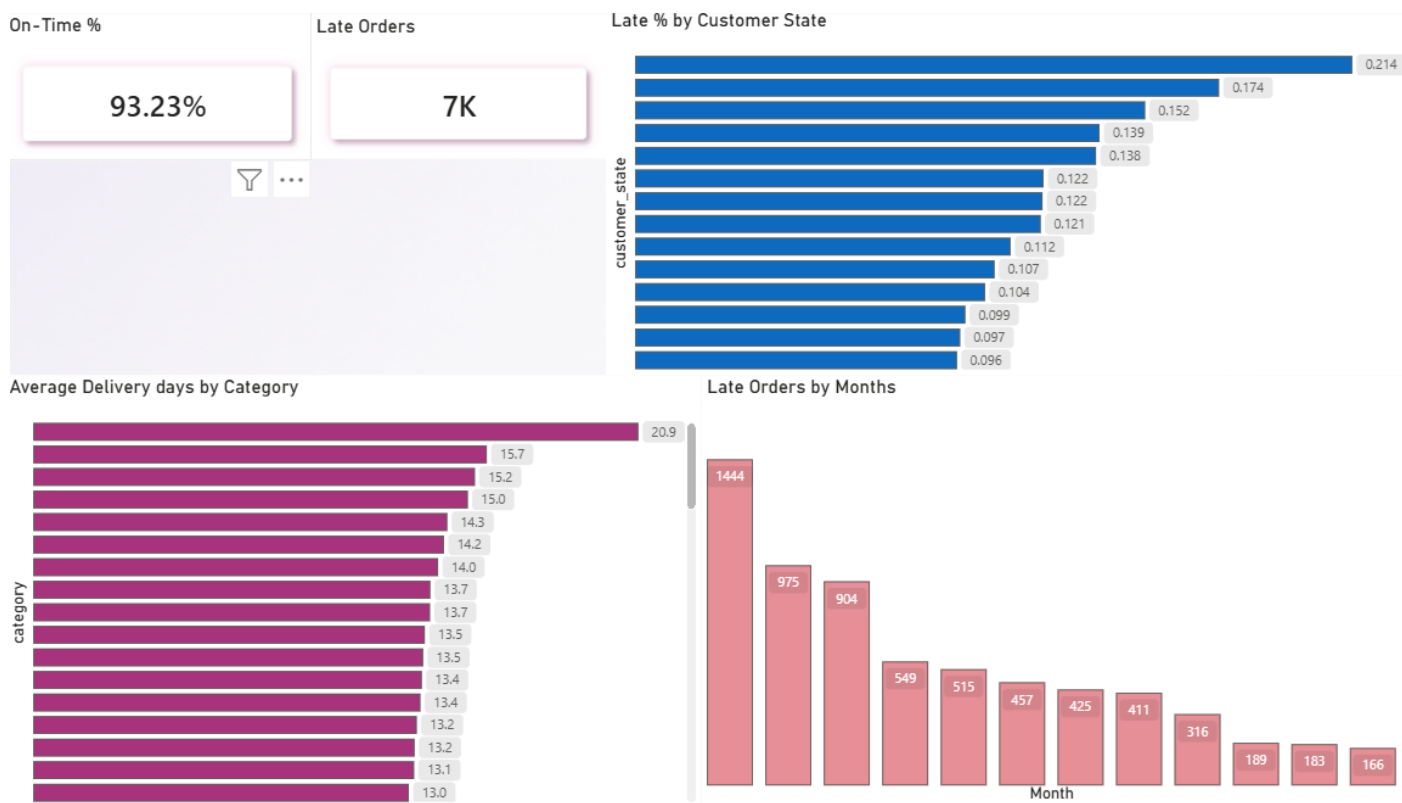


Fig1.5: Delivery & Logistics Report

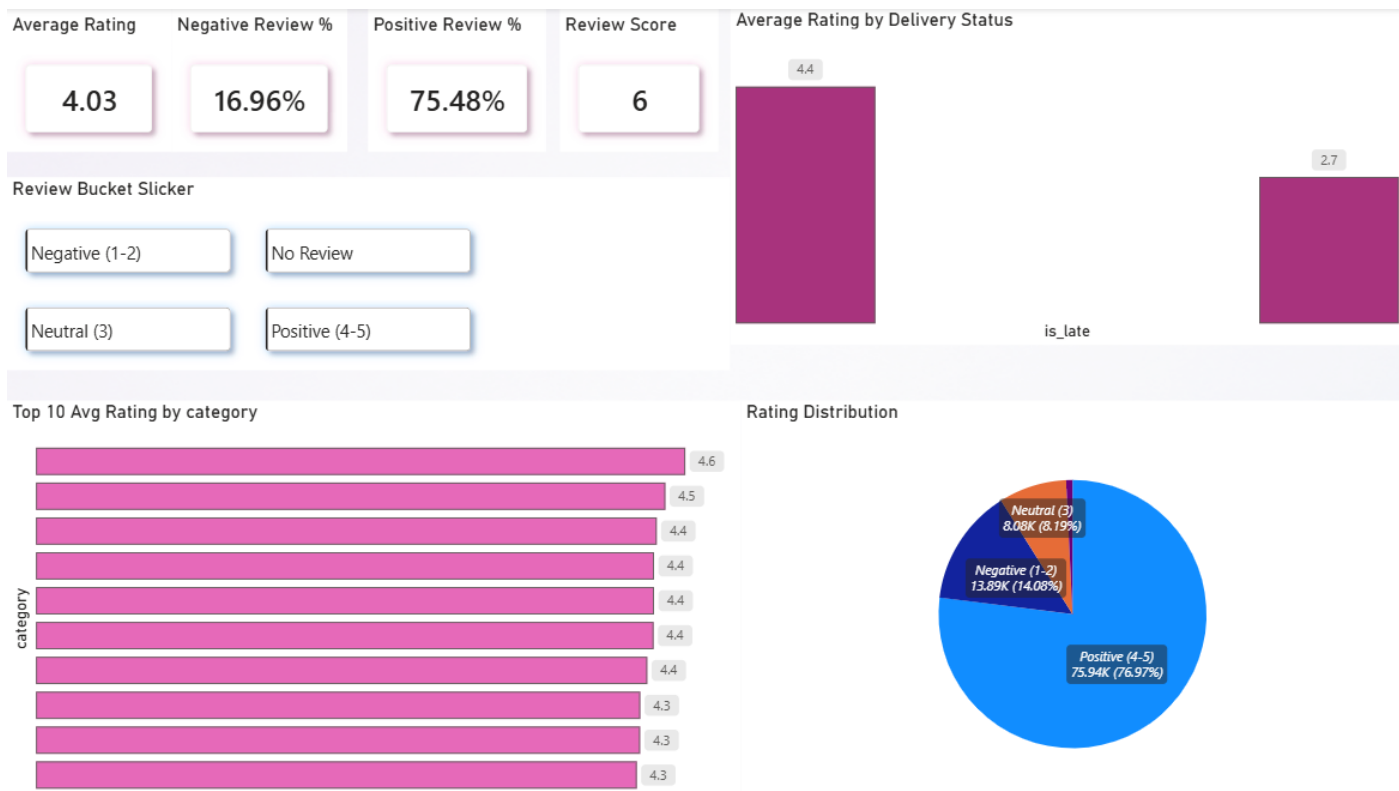


Fig 1.6: Review Report

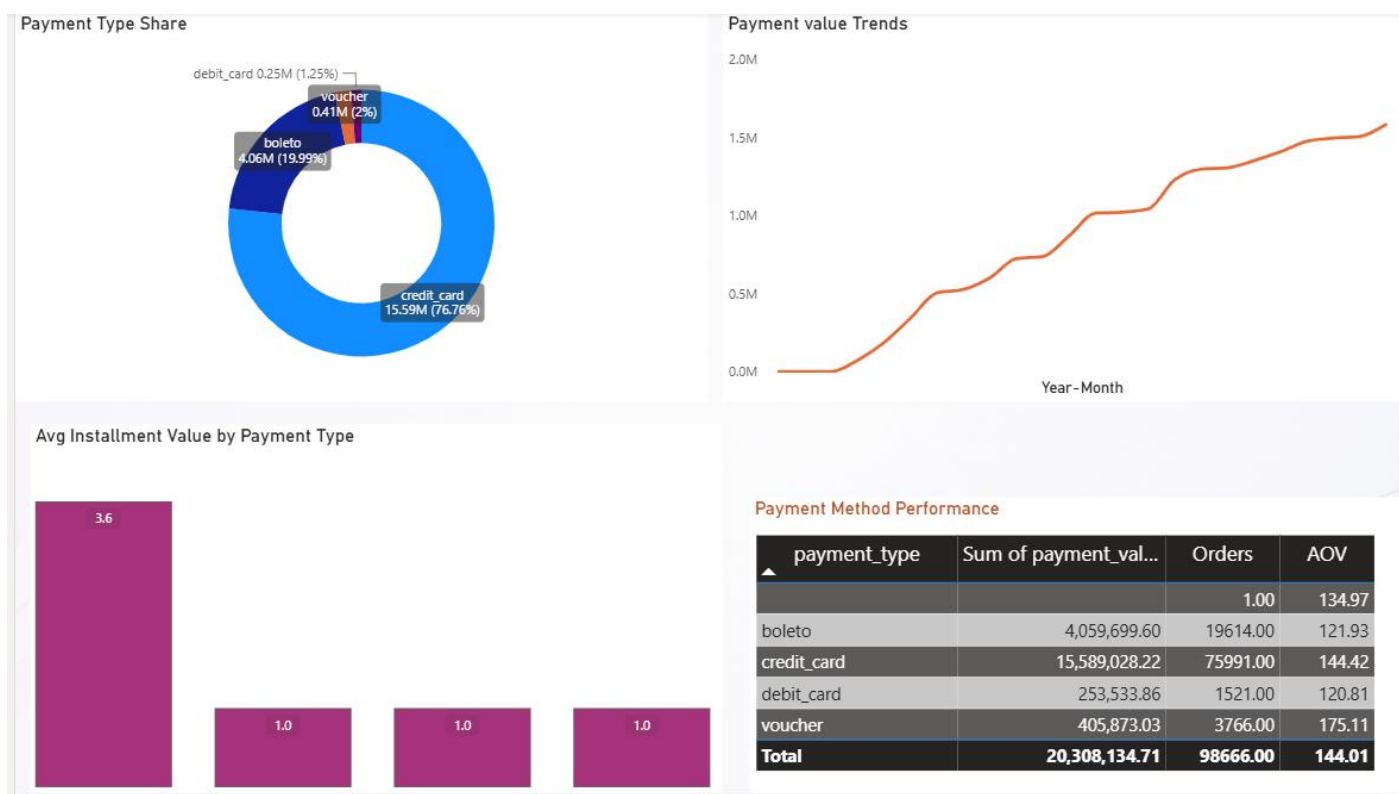


Fig 1.7: Payment Report

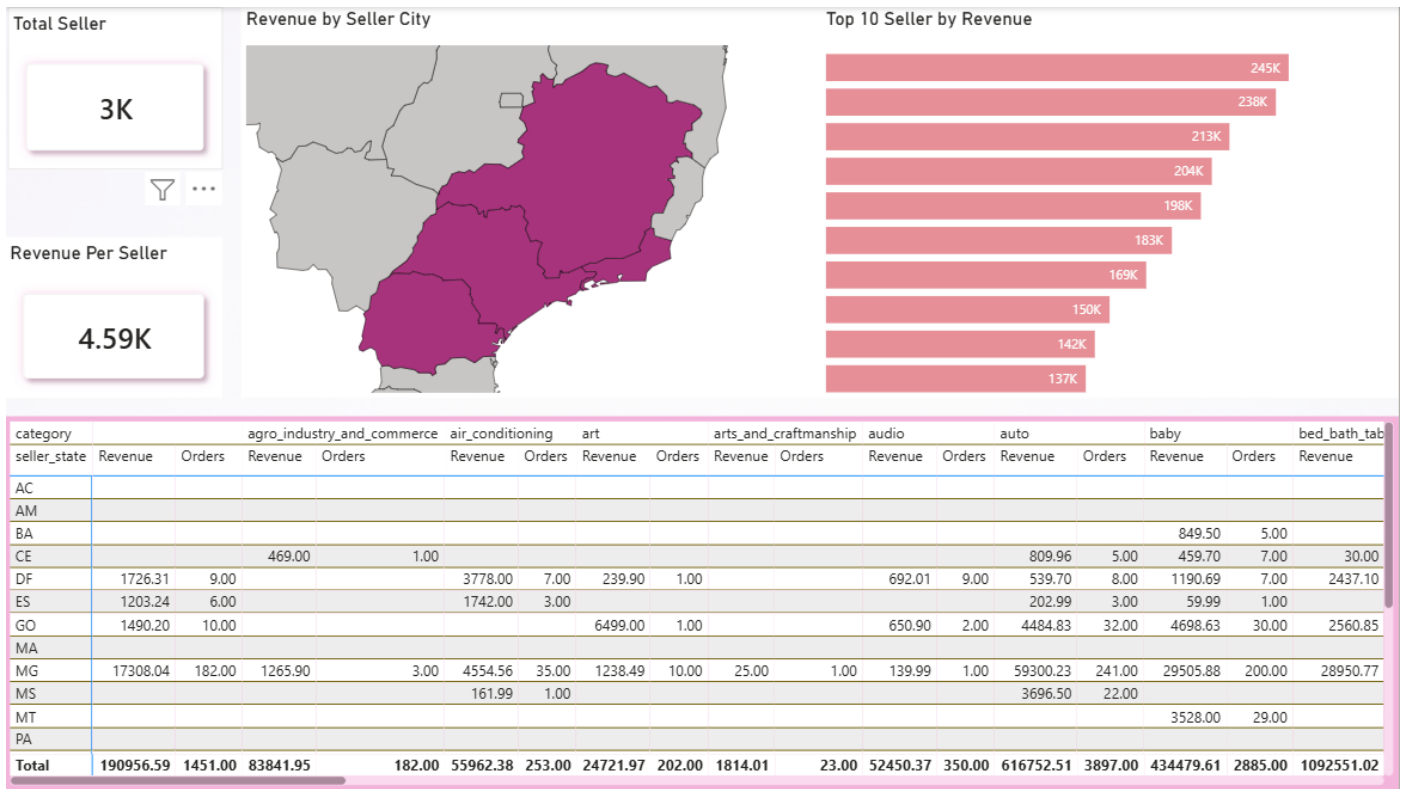


Fig 1.8: Seller Report

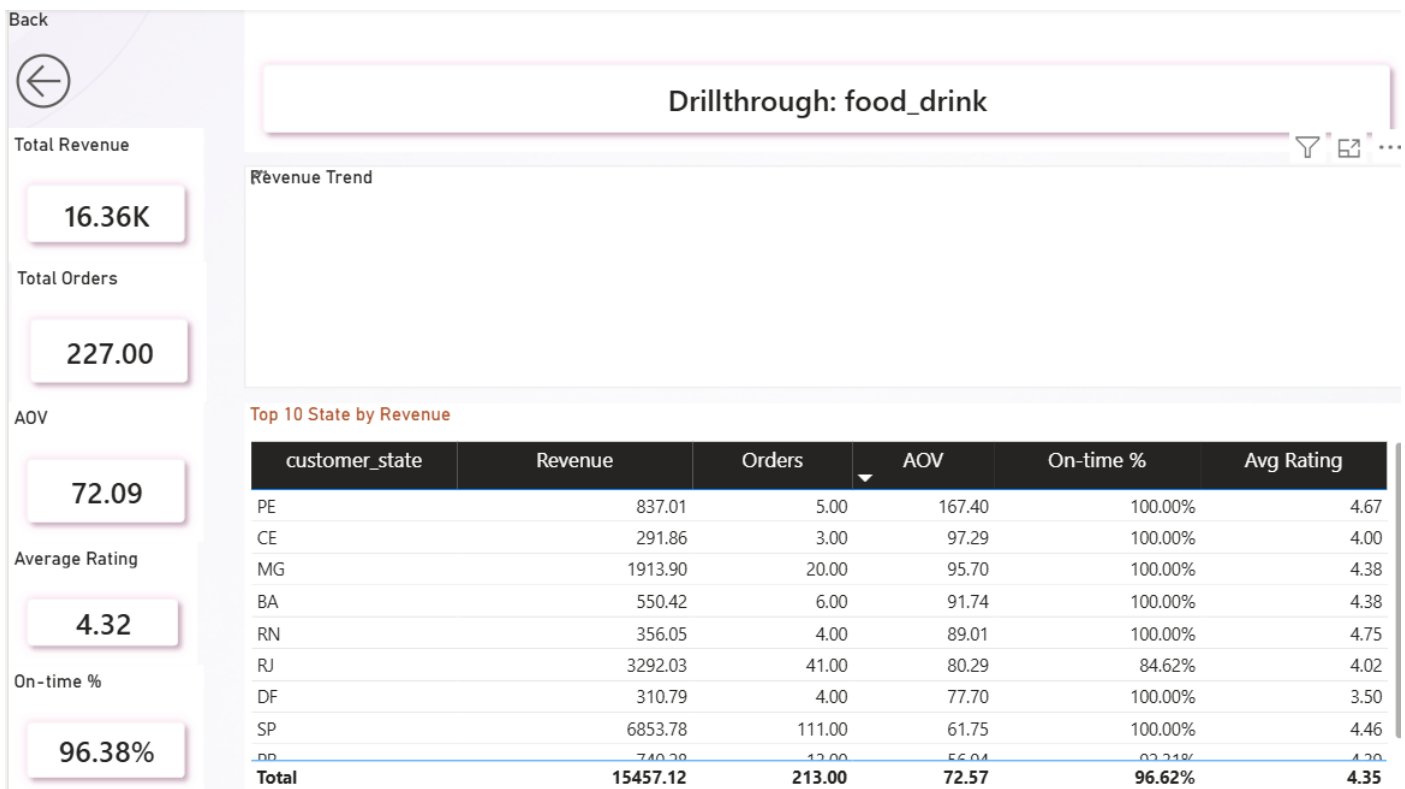


Fig 1.9: Drill Through

8. Future Enhancements

Although the current solution provides comprehensive analytical capabilities, several enhancements can further improve its value.

Real-Time Data Integration

Implement automated ETL pipelines to support near real-time reporting.

Cloud Deployment

Deploy PostgreSQL and Power BI solutions on cloud platforms such as Azure, AWS, or Google Cloud.

Predictive Analytics

Integrate Machine Learning models to forecast future sales trends and customer behavior.

Sales Forecasting

Develop forecasting models using historical sales data.

Customer Segmentation

Implement RFM (Recency, Frequency, Monetary) analysis for customer segmentation.

Customer Lifetime Value (CLV)

Calculate CLV to support retention and marketing strategies.

Churn Prediction

Identify customers likely to stop purchasing and proactively implement retention strategies.

9. Conclusion

The End-to-End E-Commerce Analytics Platform successfully demonstrates how data analytics and business intelligence technologies can transform raw transactional data into valuable business insights. By integrating PostgreSQL and Power BI, the project provides a scalable and efficient solution for analyzing sales performance, customer behavior, product trends, seller contributions, delivery operations, and customer satisfaction.

The project enables organizations to monitor key business metrics through interactive dashboards and supports informed decision-making across multiple business functions. Furthermore, the solution establishes a strong foundation for future enhancements such as predictive analytics, customer segmentation, sales forecasting, and cloud-based reporting.

Overall, the project highlights the practical application of data analytics in solving real-world business challenges and delivering measurable value to stakeholders.